

Unit - 1

Introduction to AI

Definition:-

AI (Artificial Intelligence) is a technology where machine can act like human i.e. it can be used for decision making and problem solving.

Thinking Humanly

- * The effectiveness of to make the relation between a computer and human mind which is used for Decision making and problem solving

Thinking Rationally

- * The aim is to build a computational model which is based on logic and facts.

Acting Humanly

- * The human performs certain functions as activities for problem solving and Decision making.
- * To Prove whether human is better (or) machine is better

Acting Rationally

- * The AI consist Artificial agents, the agents also consist Artifacts.

Foundation of AI:-

The artificial intelligence foundation is based on following disciplines.

1) Philosophy:-

- a) How valid Rules are used for conclusions?
- b) How does the mind arises from Brain?
- c) where does Knowledge comes from?
- d) How Knowledge leads to action?

The Philosophy is based on following factors:-

i) Rationalism:-

The Rationalism is a reasoning to understand the real world based on logic and facts.

ii) Dualism:-

The dualism refers to human body and mind where one is physical.

iii) Materialism:-

The materialism refers to the alternative of dualism where it is based on the laws of physics and physical existence.

iv) Empiricism:-

The source of knowledge in Artificial Intelligence is called empiricism.

v) Induction:-

Induction is based on general rules which are associated with elements and can be repetitive.

vi) Observation Sentence:-

The Observation Sentence is the combination of rationalism and empiricism.

vii) Confirmation theory:-

The confirmation theory is used for analysing the acquisition of knowledge from experience.

The final element in philosophy is a connection between knowledge and action.

- 2) Mathematics:- It has following Questionaries:-
- a) what are the formal rules to draw valid conclusions?
 - b) what can be computed?
 - c) How do we reason with uncertain information?

The Mathematics is based on following Disciplines:-

- i) Algorithm:- The first algorithm is Euclid's algorithm which was used for solving GCD of 2 Numbers.

* The word Algorithm comes from a Greek word Al-Khowarizmi which is step by step instruction for Problem Solving.

- ii) Incompleteness theorem:-

The Incompleteness theorem shows that formal theory is as strong as the Arithmetic statements which is based on true statement but undecidable because of no proof for the theory.

- iii) Computable:-

The fundamental of result can be interpreted by showing some functions which is infinite cannot be represented and computed.

* A Scientist by name Alan Turing has made a machine called Turing machine which is capable of computing complex problems.

- iv) Traceability:-

The Traceability is used for understanding the notation of computation along with disability.

- v) NP-Completeness:- [Non-deterministic Polynomial]

The non polynomial Completeness used for complex reasoning Problems which are known as Canonical Problems.

vi) Probability:-

Beside logic and computation the 3rd contribution of Mathematics to AI is Probability which is based on possible Outcomes.

3) Economics:-

The Economics is based on following Questionaries

- How should we make decision to maximize pay-off?
- How should we do this when others may not go along?
- How should we do this when pay off far from future?

The Economics includes following disciplines

i) Utility:-

It is based on preferred Outcome.

ii) Decision theory:-

It combines probability theory with utility theory based on a framework and uncertainty.

*Game theory:- It is a situation in Economics where it is based on rational agent used for some actions.

*Operation Research:- It is used for sequential decision problems.

4) Neuro Science:-

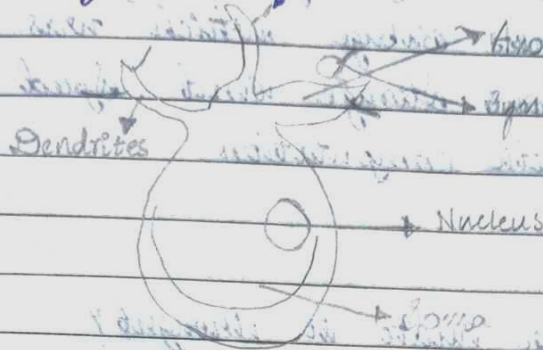
The Neuro Science is based on the following questionary

- How does the brain process information?

The Neuro Science includes following disciplines:

i) Neuron:- The neuron is a nerve cell in brain which is used for signals and the brain

includes cognitive function.



The diagram shows the nerve cell (or) neurons, each neuron consists of a cell body (or) soma that contains a nucleus. It also has many short cell branches called dendrites and it has a single long fiber called axon.

5) Psychology:-

a) How do human and animal think and

The psychology is based on behaviourism and cognitive psychology.

→ Behaviourism:- The Behaviourism is based on the study of human and animal behaviour where it should be carefully observed and controlled. It depends on human view point.

→ Cognitive psychology:- Cognitive psychology which uses the brain as an information processing device can be traced back.

6) Computer Engineering:-

a) How can we build an efficient computer?

The computer engineering is related to AI where it has 2 things: Intelligence and Artifact.

The modern digital electronic computers were invented independently and it was used during world war 2 where Alan Turing has a single purpose: deciphering German messages.

The computer science is based on generation of computer

Hardware and Speed capacity. The first computer was invented by Charles Babbage which was vacuum tube which has low storage and speed used for basic to arithmetic computation.

7) Linguistic:-

a) How does language relate to thought?

The linguistic is based on following discipline:

i) Computational Linguistics:-

The computational linguistics is a modern linguistic in AI which is based on Intersection in a hybrid field. The problem of understanding language should be more complex and it should have a context and knowledge and representation.

→ History of AI:-

1. (1943-55) - GE Station - Artificial Neurons

1 Neuron connected to another through Network very communication is possible Gates, LAN, WAN.

2. Birth of AI (1956):- Actual Implementation

* MacCarthy did some research on AI, AI should be used for problem solving.

* AI should be used for Simulation, Machine language.

3. Enthusiasm and Great Expectation (1957-1969):

* Machine should have programming tools

* Machine cannot solve unknown → Natural Intelligence (Axiom - old info)

* Physical Symbol System - machine should identify

→ Mediator

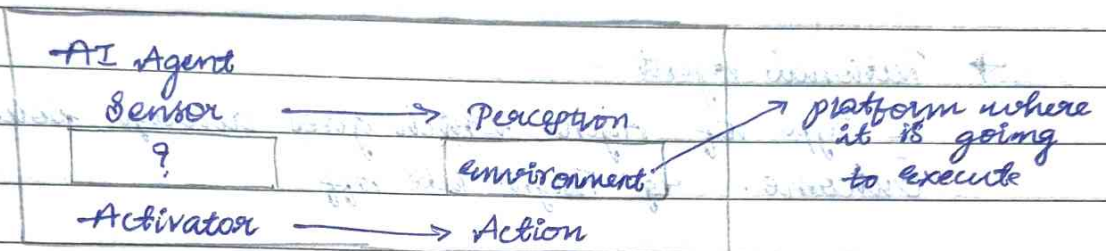
→ AI Agent and Environment:-

AI Agent is a machine/software/program consists of Sensor and Activator.

Sensor is used for Perception (Input)

Activator is used for Action

Sensor → eg:- Mobile, Door (Automatic) Light sensor [No obstacle-close]



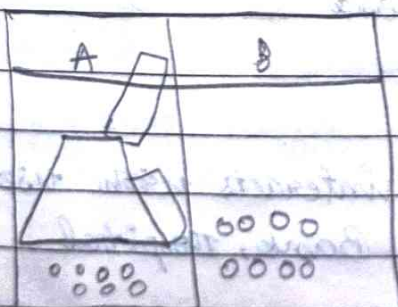
3 Types of Agents:-

1) Human Agent:- (Natural Agent)
 Sensor (Nose, ear, eyes)
 Activator (Hand, legs)
 Action → movement

2) Robotic Agent (Artificial Agent)
 Sensor (camera)
 Motor [Movement] (Circle Algorithm & other)

3) Software (or) Program Agent (Artificial Agent)
 Sensor: Chatgpt, ASCII, keystrokes - Typing/keypress, Files - ASCII value stored, Networking - connect by internet, Screen - output displayed
 (-Activator)

eg:- vacuum cleaner



Left, Right, Right → Right

Percept Sequence	Action
A[Clean]	Right
A[Dirt]	Suck
B[Dirt]	Suck
B[Clean]	Left
A[C] A[D]	(Right, Suck) Suck
B[C] B[C]	Left
A[C] B[D] B[C]	Suck

Agent function ^{Action} → (Service) can be Internal/External

Agent Program → Logic will make (C) to happen

Eg:- if (ps = clean (A)) then

move Right

else if = DCA

Suck

→ Rational Agent:-

1 Type of Agent which gives always solution (or) Outcome. Eg:- Flight Take off

Specify the Task Environment:-

* AI Agent → different task

AI Robot → class Teaching, Hospital (Surgery)

Agent Type	Performance	Environment	Activator (output) (speed)	Sensor (input)	
Taxi Driver	> Safety, comfort, maximise profit, minimise time	Road, Route, Pedestrians	Accelerator, steering, Brake	camera, Speedometer (dynamic)	(Task Journey)
Medical Diagnosis System	Healthy patient, Reduce cost	Hospital, Staff, Patient	Questionnaires, Test, Treatment	Symptoms, Keyboard Entry	
Robotic Agent	Separating, packaging, Dispatch	Company, packaging unit	Artificial Hand	camera	

→ Soft Bot:- AI Software agent it interacts with users.
different types based on News, Bank, Hospital

→ properties of Task Environment:-

I 1) Fully observable:- Sensor of AI Agent should give access for all states to its environment.

Eg:- 1) car → Route - Source, Intermediate, Destination
Driver

2) Partially observable:- where all states are not accessible only few states are accessible.

Eg:- Medical → patient → Test → Blood → lab
Diagnosis System Test

3) Unobservable:- where there is no sensor

Eg:- Human Taxi driver doesn't need camera

II 4) Single Agent:- For solving problem - solution one agent is enough

Eg:- cross word puzzle

S	D	C
	O	A
	G	T

5) Multi Agent:- For problem solving minimum 2 agent is required.

Eg:- Chess Board: program Agent (computer) & human

III 6) Compete:- where 1 agent compete with another agent

Eg:- chess

7) co-Operative:- 1 agent will support another agent

Eg:- 1 taxi driver should support another by following Traffic Rules

IV 8) **Deterministic**:- Next State Action depends on current state
eg- If Symptom is there then only test is required then Treatment is provided.

unhealthy (current state) → Healthy (Next state)

9) **Stochastic**:- Next State Action doesnot depend on Current State.

Eg:- Taxi Driver → 1 Taxi Driver doesn't depend on another Taxi Driver

1 Taxi → AI another Taxi → Human

10) **Non Deterministic**:- It is where action is based on Possible Outcome (solution)

eg:- Route urban → fast Driving

Route Rural → slow Driving (Narrow, Animal, pedestrians)

V 11) **Static**:- Environment is fixed

eg:- Medical Diagnosis System (1 hospital visit.)

12) **Dynamic**:- Environment changes

eg:- Route $\begin{cases} \text{urban} \\ \text{Rural} \end{cases}$ Alternative

13) **Semi-Static** - Environment is fixed but Score of Agent (or) performance gets changed.

eg:- chess score

VI 14) **Continuous** - where Action will be without Interval

eg:- Taxi Driver → until passenger stop they won't

15) **Discrete** - Interval in percept (or) Input

eg:- Chess board, crossword puzzle.

VII 16) episodial - It is stochastic where next state doesnot depend on current state.
Eg:- weather Forecasting → kela weather not depend on Bngla

17) Sequential - It is Deterministic where next state (action) depends on current state (percept)
Eg:- chess, crossword puzzle.

Task Environment	Observable	Agent	Deterministic	Static	Episodial/ Sequential	Discrete/ continuous
crossword puzzle	Fully Observable	Single	Deterministic	Static	Sequential	Discrete
chess with clock	Fully Observable	Multi Agent	Deterministic	Semi Static	Discrete Sequential	Discrete
Taxi Driver	Partially Observable	multi Agent	Stochastic	Dynamic	Sequential	continuous
medical Diagnosis system	partially Observable	Single	Stochastic	Dynamic	Sequential	continuous

→ Structure of AI Agent:-
(Architecture) Skeleton/Blueprint

Agent = Architecture + Program (Agent Program) ^{function}

- > Agent Program
- > Agent Function — Percept (Sensor)
Action (Activator)

The Agent does the function with help of Percept and returns Action with help of Activator

Percept → can be based on current state / single percept
Sequence of state / history is Required

→ Table Driven Agent: Has Table, Rows and Columns
Table Driven Agent (percepts) returns Action
Percept → Sequence of Input - Initially empty
Table → Set of Actions based on Percept
where Table should be fully specified.

(Add) Append percept to end

(Action) ← Lookup Action (percept, table)
Return Action

Imp → Types of AI Agent:-

- 1) Simple Reflex Agent
- 2) Model Based Agent
- 3) Goal Based Agent
- 4) Utility Based Agent
- 5) Learning Agent

1) Simple Reflex Agent:- It is one type of AI Agent
it is based on current Percept & current state

* Next state of Action depends on current state

* Does not bother about percept sequence

eg:- vacuum cleaner (Suck, Move) → Reflex

Simple - vacuum Reflex (location, status)

(condition) If status = dirt then

(Action) Suck

else if location = A and status = clean
move Right

else if Location = B and Status = clean
move Left

Simple Reflex Agent

AI

Percept

Agent

Sensor

How the world is
look like Now?

Environment

condition based
action Rules

Action

Activator

what Action I
should Do Now?

→ Algorithm:-

Simple - Reflex - Agent (Percepts) return action

Persistent - condition Based Action Rules

STATE - Interpret - Input (percept)

RULE - Matching (State, Rule)

ACTION - Rule. Action

Return action

Based
on Rule

Vacuum cleaner - It is based on local current state, has knowledge only about dirt then it only Sucks (Action) and clean, where move cannot be performed. So flip is situation where either dirt or clean will be there either 1 will be performed.

2) Model Based Agent:-

It is one type of AI Agent which is based on sequence of percept. It should maintain Internal State where history is required through sensor.
eg:- vacuum cleaner

SEE + REMEMBER → DECIDE

- * In vacuum the move has 2 sequence either Left (or) right
- * Taxi-camera-Front car applied brake-AI agent stop next car should also Apply brake.

Model Based Agent

AI Agent

Percept
Sensor

State

How the world
looks like now?

Environment

Which state
will evolve?

What my Action
will do?

Condition based
Action Rules

Action

Activator

Highway
Service
Roads

What Action I
should do Now?

eg: changing lane 4 lanes, Steer
2) move steer (left, right)

→ Algorithm :-

What it feels

What it does

I do → clean

Model - Reflex - Agent (Percepts) return Action

uses
Rules

Persistent (State, Model, Rule, Action)

what agent thinks
world looks like
understanding
How things

STATE - current perception of world state

Room is dirty

I am at loc A

Clean → Room clean

Move → Room another

MODEL - How the Next State depends on Current State

RULE - Based on Matching (condition Based Action Rule)

ACTION - Set of Actions based on Initial values

STATE - Update - state (State, percept, Model, Action)

RULE - Matching (States, Rule)

ACTION - Rule: Action

Return Action:

1) Looks at I/p (what it sees) & gives an action (what it does)

2) Rules are always stored & used

3) what the agent thinks the world looks like Right Now?

4) understanding & things & "if I do this → ? (Happen Next?)"

Action

5) Agent checks Rules & picks Action

6) updates State using → Past memory, current I/p, Model (understanding)

7) Agent Lo → "1 Rule matches (current situation?)"

STATE → clean
Rule → Move

8) After finding Rule → do that Action

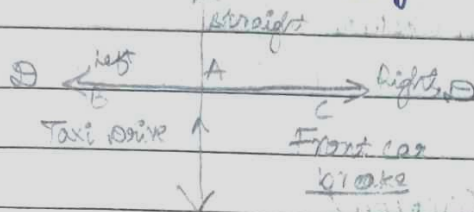
9) finally Agent Vb → x

3) Goal Based Agent :- Should be a predetermined goal to be assigned to agent.

* It is more flexible than Model Based Agent.

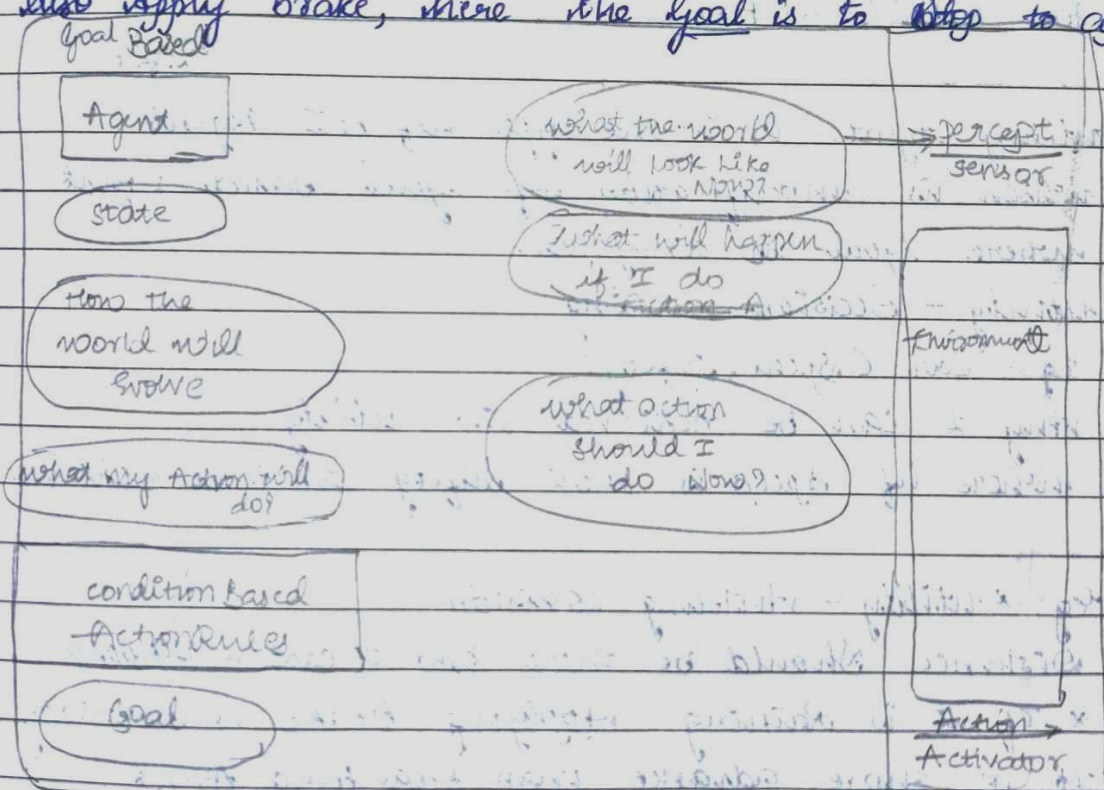
* Whether there is a goal (or) target

HB:- Happy (Successful) and unhappy



* Goal is either Left, Straight (or) Right

* If the front car applies brake the Taxi Driver should also apply brake, where the goal is to stop to apply brake



Group of components

Algorithm:-

Goal-based Agent (percepts) return Action

Persistent:-

STATE - Agent's current Perception of the world

MODEL - Is based on Next State of Action with current State

GOAL - Desired Goal with current State

RULE -

ACTION -

State - update.State (State, rules)
 Model - (State, Action, percept, Model)
 Rule = Matching (State, rule)
 Action - Rule. Action
 if - goal = test (State, goal) then = false then
 creating ← Formulate - Problem (State, goal
 Solution (Problem)
 else
 No Action = first (action)
 else
 No Action
 Return
 For Test (BFS, DFS)

4) utility Agent:- It is one type of AI Agent
 which is modification of goal based Agent
 where goal is assigned.

* utility - Decision making

Eg:- Car (Speed, Safety)

Any 1 can be achieved in utility

where if Speed is there Safety can't be achieved

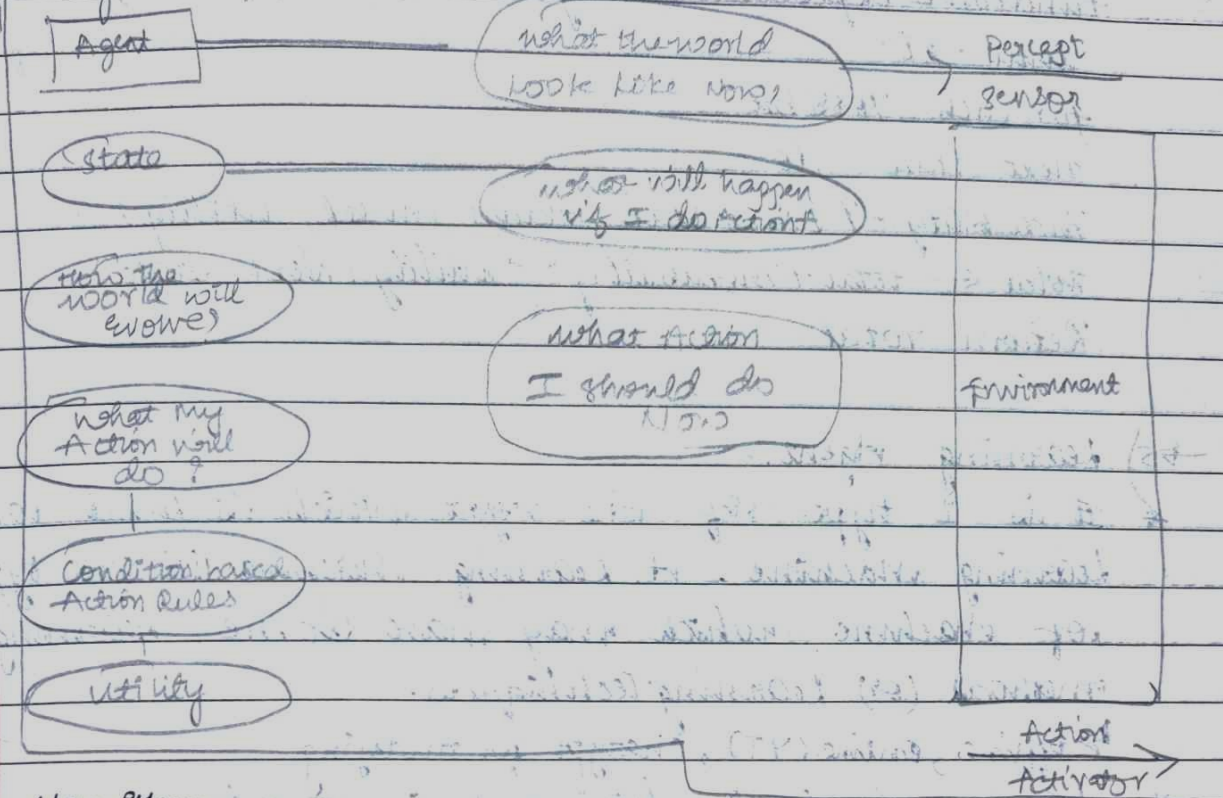
eg:- * utility - Avoiding collision

Distance should be there b/w 1 car to another

* If it is Raining Applying brake is Necessary

* It is more advance than goal based Agent

Utility based Agent



PART-C

Algorithm

utility - based - Agent (percept) return Action

State - Agent's current Perception of the world

Model - Next State of Action is based on current State

utility - Decision making Function

State - update - State (State, Rule)

Action - action (State)

Shortest path

← best - action - null

Max - utility → ∞ Infinity

for each a in action do

Expected - utility - Expected utility (action, state, Model, utility)

If expected utility $\geq$ Max utility then

Max utility = Expected utility finite Result

best Action ← a Shortest Distance

end if

end for

Return action (best Action)

Group of Elements
of Object,
Interaction
(Action)

Function - expected - utility (Action, state, model, utility)
total $\rightarrow 0$

Safety / Speed

for each possible
next state do

possibility
of Best Action
maximized
utility

Probability - $P(\text{Next state, Action, model, utility})$

total $\leftarrow$ total + Probability $\rightarrow$ utility (Next state)

Return total

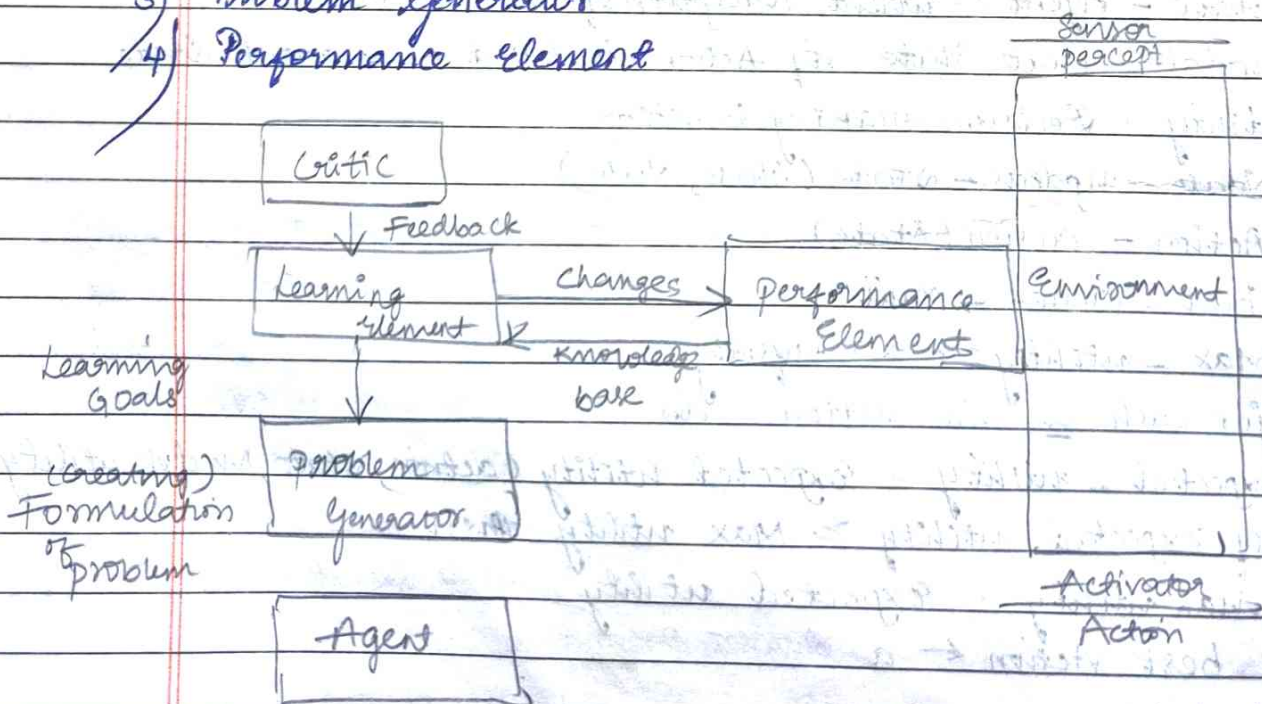
→ 5) Learning Agent:-

* It is 1 type of AI Agent which is based on Learning machine. A Learning machine is one type of machine which may have some learning methods (or) Learning Techniques.

→ offline, online (YT), chatgpt for studying

* Learning Agent is having 4 components

- 1) Critic
- 2) Learning element
- 3) Problem generator
- 4) Performance element



- 1) Critic is a component of Learning Agent which is used for feedback from ^{another} agent. (update)
- 2) Learning is a component of Learning Agent which is used for achieving Learning Goals (aim)
- 3) Problem Generator is a component of Learning Agent which is used for formulation of problem (Designing)
- 4) Performance element is a component of Learning Agent which is used for Best Performance & New Information, New Experience. (Different environment)
 - * changes - Learning Element makes changes for Performance
 - eg: element (If you learn more, performance will be also more & vice versa) (update $\rightarrow$ state change)
 - * changes which are taking part that should be stored in Knowledge Base (stored info for L Agent)
 - * environment consist 2 components
 - 1) Sensor $\rightarrow$ Percept
 - 2) Activator - Action.

eg:- Taxi Driver - Automated

Taxi Driver has changed the lanes without the Indicator it has gone left side, where human Driver will be there that driver will use a ~~bad~~ word for this [feedback

$\rightarrow$ will be given to Automatic Taxi Driver] $\rightarrow$ Critic

(Current TD)

* So Automatic Taxi Driver will learn from this which is Learning element and it will use Next time Indicator, this will be stored in Knowledge Base. without Indicator - Bad Performance with Indicator - Learning

* Taxi Driver Reaching Destination - Problem Generator

* Learning Goal - Safe Journey with Indicator

Eg 2:- Taxi Driver - If it Rains and Road is wet then the car should move slow - Learning Element, Deceleration - Driver will apply often brakes where Speed becomes Slow
If car Jumps, Passenger will tell AI Agent to Reduce Speed - Critic - feedback
Learning goal - safe Journey

Eg 3:- Chess - checkmate

In a chess Problem Learning goal - Checkmate, the AI Agent should make best Performance
Goal depends on Performance

Horse should move in L shape it will achieve goal, performance is Good

If he moves in I shape, performance will be bad, achieved goal cannot be Achieved

Algorithm:- / Pseudo code

Learning Agent (Percepts) return Action

STATE - current state of world

Goal - Goal of Learning element

Initialize -

1) Performance element

2) Critic

3) Learning Element

4) Problem Generator

5) Knowledge Base

Loop Forever

I. Perceive Environment (Percept)

II. State - Update State (state, percept)

III. Action (State) (Execution)
 moving

IV. Execution

Move Stop Rain No

V Feedback-Critic (State, Percept)

VI Learning Element - (Feedback)